

FABLE-Pakistan: Technical Decisions and Methodology Log

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Purpose of This Document

The main project report explains what the FABLE-Pakistan pipeline is, how it was built, and what it produces. This document goes one level deeper: it records every non-obvious decision made along the way — unit conversions, tariff-code corrections, conversion-factor sourcing, and the specific data-quality problems found, both the ones that were fixed and the ones that were investigated and deliberately left as-is. Figures and sources quoted below are quoted exactly, so this can serve as an audit trail for anyone extending or questioning the pipeline later.

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1 Production Dataset Decisions

1.1 Unit standardisation

Cotton lint was originally reported in thousands of bales, Pakistan’s traditional trading unit. It was converted to thousands of tonnes using the Pakistan Central Cotton Committee’s standard bale weight of 170 kg, so 9,000 thousand bales becomes $9,000 \times 0.170 = 1,530$ thousand tonnes.

Eggs required more thought. Pakistan’s statistics report egg production as a count, and an earlier pass kept it that way on the reasoning that eggs are counted, not weighed. That doesn’t hold up: FABLE’s balance table works in 1,000-tonne units for every commodity, and the trade side reports eggs by weight, so a count can’t be joined against either without converting somewhere. Better to convert once, at the production stage, than leave it as a trap for whoever builds the join. The conversion uses an average hen-egg weight of 47 grams, from the same FAO Technical Conversion Factors table that supplied the livestock carcass ratios elsewhere in the pipeline (58 grams for non-hen eggs, not used here since production is overwhelmingly hen eggs). A related figure in the same table — that roughly 5.7% of production goes to hatching rather than food — was not applied, since that is a utilisation split, not a unit conversion.

The most consequential unit fix was a labelling bug affecting three sheets: **Sunflower**, **Jute**, and **Other Oils**. It surfaced as a visibly impossible spike while building an early exploratory chart — Sunflower’s national total ran as high as 600,000 thousand tonnes in a single year, which would put Pakistan’s sunflower harvest in the hundreds of millions of tonnes. The cause: these three sheets’ own title rows read “*Production (Tonnes (raw))*”, not the “*000 Tonnes*” convention every other sheet uses, and the standardisation script’s unit-detection logic only ever checked for the patterns ’000, Bale, and Million, so these three were skipped rather than divided by 1,000. Once fixed, Sunflower’s total moved from an implausible 80,000–600,000 range down to a plausible 80.6–603.9 thousand tonnes. The fix was tightened once to avoid a false positive on “Raw animal fat”, whose name also contains the word “raw” but which was already correctly labelled.

The “no data” convention. Every genuinely missing year is written as the literal string “no data”, never interpolated or estimated. A dash in a source table, by contrast, is treated as a real reported “negligible or nil” figure and converted to a numeric 0, since that is what the publication is actually asserting.

Province Sum Check. Every sheet carries a column flagging whether the four provincial figures sum to the national total within a 0.5-thousand-tonne tolerance for source rounding. Known pre-existing mismatches — Rapeseed (2010-11, 2014-15, 2018-19), Other oils (2011-12), Other pulses (2014-15, 2016-17), and Tobacco (2014-15) — are flagged by this column and were not corrected, since they are errors in the original publications, not in this pipeline.

1.2 Aggregating Pakistani crops into FABLE’s composite commodities

Several FABLE commodities are explicit composites, built by summing distinct Pakistani crops:

- **Rice (paddy)** = Basmati + IRRI (ASP’s own total already sums both).
- **Lentil/Beans** = Masoor + Mash.
- **Other pulses** = Guarseed + Mattar on the production side (the trade side treats guar differently — Section 3).
- **Other oils** = Linseed + Safflower + Castorseed.
- **Nuts** = Almond (ASP) + Walnut (FVC, KPK) + Pistachio (FVC, Balochistan).
- **Other fruits** = the ASP base group (guava, peach, apricot, plum, pomegranate, pear) plus every FVC fruit not already assigned its own slot (melons, ber, loquat, papaya, chikoo, strawberry, fig, cherry, mulberry, phalsa, jaman, persimmon, litchi, olive).

- **Other vegetables** = the FVC vegetable total minus tomato, sweet potato, peas and sugar beet, which each have their own slot.
- **Other spices** = coriander + turmeric + ginger (ginger only from the 2023-24 FVC edition, Sindh only, and negligible in size).

When summing, a crop missing for a year it should have data for is treated as zero only if at least one other crop in the same group has real data that year; if none do, the whole composite cell is left as “no data”.

1.3 Cross-check versus overwrite: integrating five FVC editions

FVC is published in five-year windows, five editions in total (2008-09, 2012-13, 2014-15, 2018-19, 2023-24), each verified from the tables’ own column headers rather than the filename. Eight crops — apple, banana, date, grape, onion, tomato, potato, and citrus — appear in both ASP/PBS and FVC. For these, every new FVC edition was used only to *cross-check* the existing figures, never to overwrite them, because later FVC editions periodically fold several named crops into one combined “Others*” line for a specific province — the 2023-24 edition’s Punjab fruit table, for instance, bundles apple, almond, walnut, papaya, ber and loquat into one figure, making that edition’s own apple value for Punjab read as zero even though Punjab obviously still grows apples. Treating a newer edition as automatically more authoritative would have overwritten a clean, gap-free ASP series with this kind of artifact. Genuine FVC-only commodities (peas, sweet potato, coconut, the composite aggregates) carried no such risk and were filled or extended directly.

Extraction used `pdfplumber` with position-based column matching: every number on a page is assigned to the nearest year column by horizontal position and the nearest crop label by vertical position, rather than assumed to appear in printed reading order — a choice that caught a real defect, described below.

1.4 Closing the year gaps

- The single interior gap in the whole 25-year series, fiscal year 2009-10, was closed using the FVC 2012-13 edition (sourced from the Lahore School of Economics repository, since the ministry’s own site no longer hosts it) — the only edition obtained that covers that year.
- Sunflower was completed back to 2000-01 using **PBS ASP 2010-11, Table 32**, a source already sitting in the project folder. The sheet’s own note had already claimed this table as a source for those years, while the cells themselves read “no data” — cited but never extracted from. Two of the recovered years, 2000-01 and 2005-06, are FABLE calibration years.
- Sugar raw was extended from 2011-12 through 2023-24 using MINFAL ASP 2023-24’s own sugar table, again a source whose absence had been assumed rather than checked.
- Livestock and fish were extended through 2023-24 from the same ASP 2023-24 edition.

The recurring lesson: before concluding a gap is unsourceable, check whether the source already cited for it has actually been read.

1.5 Data-quality findings on the production side

Confirmed against an external event. Cotton lint’s national total crashes to 834.7 thousand tonnes in 2022-23 (from a normal 1,400–1,740 range) and Balochistan’s date production crashes from 595.0 to 1.45 thousand tonnes the same year before rebounding to 660.4 — both match the well-documented 2022 Pakistan floods, which struck ahead of harvest, destroyed roughly 90% of the date crop in Balochistan’s Makran belt, and led the International Cotton Advisory Committee to cut its Pakistan cotton estimate by a third. Balochistan’s rice production shows the same crash-and-rebound in 2010-11, 2012-13, and 2022-23, each matching a separately

documented flood in the Nasirabad/Jaffarabad rice belt. Punjab’s sesame and rapeseed output rises sharply beginning precisely in 2019-20, matching Pakistan’s Oilseed Enhancement Program, a subsidy scheme launched that year to cut the edible-oil import bill.

Plausible but not independently confirmed. Punjab’s Gram/Chickpea dips sharply in 2011-12, matching the trough-and-recovery pattern agronomy research associates with its largely-rainfed growing region, though 2011-12 rainfall specifically couldn’t be confirmed. Punjab’s Sugarbeet output starts from zero in 2009-10 and grows steadily after, lining up with published field-trial timing for autumn sugarbeet varieties introduced around then, without a source confirming that exact year. Both left as reported.

Real transcription and formatting defects, handled rather than propagated. A district-wise Sorghum table had been misread in an earlier pass, writing Punjab’s 2023-24 total into the 2024-25 cell and an unrelated recovery figure into 2023-24; corrected directly against the source table’s own printed province totals. A Balochistan Rabi-season tomato table in the FVC 2012-13 edition prints only four numbers across five year columns (2009-10 is genuinely blank in the source), and `pdfplumber` emits those four numbers left-packed — a naive left-to-right reading would have written 2008-09’s figure into 2009-10 and shifted every later year over by one. Position-based extraction avoided this; the table’s own Total row, which nets to zero once 2009-10 is correctly left blank, confirmed the diagnosis. The same edition’s Sindh mango figure for 2009-10 is a roughly ten-times transcription error in the source itself (37,897 against a true figure near 379,000); this couldn’t propagate into the Other Fruits aggregate, since mango is subtracted out of both the province total and the crop list before the aggregate is computed, so the error cancels.

Pakistan’s sugar tables carried several defects across editions: the relevant table is renumbered between editions (Table 164 becomes Table 163), so it has to be located by caption, not number; the 2016-17 row straddles a page break onto a continuation page with no caption at all; and that row’s printed Total omits the sugar-produced figure, printing five numbers where a complete row has six — reading the fourth number blindly would have mistaken the recovery percentage for national output, so the national figure was instead derived by summing the four provinces and cross-checked against the prior edition, which agrees almost exactly. Separately, the 2023-24 sugar table is an exact repeat of 2021-22’s figures — almost certainly a copy-paste error in the source — and the fish table marks 2023-24 with a symbol whose own footnote says the value was repeated, not newly measured. Both were kept exactly as published and flagged in the relevant sheet’s notes.

Two commodities carry a visible step where a later ASP edition revised an earlier year without correction: Eggs’ 2021-22 figure moved from 21,285 to 22,512 million eggs (+5.8%) and Poultry meat’s from 1,809 to 1,977 thousand tonnes (+9.3%) between editions. The earlier values were kept, not overwritten, so both series carry a visible discontinuity at the boundary — for Eggs, roughly 58 thousand tonnes once converted, larger than the underlying year-on-year growth itself.

The Milk series’ convention — gross production versus net of on-farm consumption — was confirmed by finding an edition that prints both columns side by side: the existing series matches the gross-production column exactly across all four years checked.

2 Trade Concordance and Conversion-Factor Decisions

2.1 Structure

The concordance table records, for every HS code that might belong to a FABLE commodity, an action — `INCLUDE` (query as-is), `CONVERT` (query, then multiply by a factor), `EXCLUDE` (never query — manufactured product, double-count risk, or wrong slot), or `ZERO` (no meaningful trade) — plus the matching FAO item for validation, and for `CONVERT` rows, the factor and its source.

It grew from 222 to 530 rows over the project, mostly because Comtrade codes fish species by species while FABLE has one Fish commodity; kept as a deliberate trade-off rather than collapsed into an unreliable rollup. As of the final session it holds 435 **INCLUDE**, 49 **CONVERT**, 31 **EXCLUDE**, and 15 **ZERO** rows. Ten HS codes deliberately appear on two rows — one where the code is genuinely queried, one **EXCLUDE** row under a different commodity explaining where it actually belongs — and every rebuild checks automatically that no code is ever queried under more than one commodity at once.

2.2 Pulling raw trade data

Comtrade allows at most twelve periods per call and imposes a URL-length limit, so the pull chunks both years and HS codes into batches. Pakistan has a genuine reporter submission for every year from 2003 onward; only 2000–2002 are true full-year gaps, filled instead with partner-mirror data (what every other country reports trading with Pakistan, direction flipped and summed across every partner). Every row carries an explicit **Data Source** tag — *Reporter*, *Reporter (zero trade reported)*, *Partner-Mirror*, or *no data* — so a genuine zero-trade year is never confused with a year Pakistan simply didn’t report, a distinction that mattered a great deal later (Section 3).

2.3 A placeholder-code bug on the first pull

The first pull returned zero trade in every year for twelve of its 177 codes, including commodities Pakistan clearly trades — fish, shrimp, groundnuts, coconut oil, cotton waste. These turned out not to be real Comtrade codes: they were four-digit tariff *headings* with a placeholder .00 suffix (e.g. 0302.00 for “fish, fresh or chilled”), when Comtrade only records trade at the real six-digit codes underneath. All twelve were expanded into their real leaf codes and re-pulled; every one showed real, multi-million-dollar trade once queried correctly.

2.4 Conversion-factor corrections

Item	Factor	What changed and why
Cocoa, defatted paste (HS 1803.20)	1.25 → 1.18	Had matched non-defatted paste on the assumption that removing fat needs more beans per tonne. ICCO’s own bean-equivalent factors group defatted paste with cocoa powder/cake at 1.18, not with non-defatted paste.
Rice, husked/brown (HS 1006.20)	1.25 (validated)	IRRI’s published milling figures confirm the 80% husking yield the factor already assumed; the number was right, only the citation was missing.
Whole milk powder (HS 0402.21)	7.70 (moderately sourced)	No single FAO table gives this directly, but independent dairy-processing references converge on the same 1:7–8 reconstitution ratio, cross-checked against Codex’s compositional standard.
Dried onion (HS 0712.20)	EXCLUDE → CONVERT, 9.00	Had been excluded purely for lack of a factor, not for a double-counting reason. USDA Agricultural Handbook 697’s dehydrated-vegetable table gives a farm-weight equivalence of 9.0.

Item	Factor	What changed and why
Groundnut, shelled (HS 1202.42)	restructured, 1.25	Two HS rows were mislabelled (“seed”/“in shell” swapped with their actual Comtrade meaning), meaning shelled groundnuts were summed with no shelling adjustment at all. Restructured into correct in-shell and shelled groups.
Live cattle, sheep, goats	carcass wt. per animal (181/19/16 kg)	Comtrade reports these as a head-count, not a weight, so a kg-to-kg carcass ratio can’t apply directly. The same FAO table that supplied the ratios also publishes an average carcass weight per animal, letting head-count convert to tonnes directly without double-converting. Effect is small: +29.5 kt to Beef and +4.3 kt to Goat & Lamb exports across 26 years, under 0.1% of production either way.
Guar gum (HS 1302.32)	1.00 → 3.45	Had been left at raw-gum weight since the earliest trade work, flagged repeatedly as unsourced. FAO/JECFA’s guar composition assessment plus two independent guar-processing industry sources converge on a commercial refined yield of 28–30% of seed weight; midpoint 29% gives a reverse factor of 3.45.

Table 1: Conversion-factor corrections made over the course of the project.

3 HS-Code Revisions: the Project’s Largest Source of Hidden Data Loss

The Harmonized System is periodically revised — in 2002, 2012, 2017 and 2022 across the years this project covers — and old codes are retired and split at each revision. Comtrade answers a query in whatever vintage was in force that year, so a retired code returns normal-looking trade right up to the revision and an exact zero after it, which looks exactly like real absence of trade. This turned out to be the single most dangerous failure mode in the whole dataset.

3.1 Citrus (kino), 2017

The concordance mapped kino, Pakistan’s largest fruit export, to HS 0805.20. Correct when written, but the HS 2017 revision retired it and split it three ways: 0805.21 (mandarins), 0805.22 (clementines), 0805.29 (tangelos). The concordance was never updated, so Orange/Citrus exports appeared to collapse from 2017 onward — \$2.82M against a true \$76.77M for 2023 alone, and roughly 300,000 tonnes a year of kino vanished from the record. This escaped an earlier audit because that audit’s test was “does this code return zero in *every* year,” and 0805.20 had seventeen genuinely good years first. Verified before fixing: 0805.20 and its three successors are never simultaneously non-zero, so the retired code was kept for the years it covers and the successors added alongside, restoring one continuous series with no double-count.

3.2 Nuts: the same bug in both directions

Generalising the citrus fix into a proper test — for every code that drops to zero and stays there, does a sibling code in the same heading carry real trade nothing else is querying? — immediately

found the same bug in Nuts, in both temporal directions at once. HS 2022 retired 0802.90 (“other nuts n.e.c.”) and split it into pine-nut and residual codes; the concordance’s own note on the retired code had explicitly said it was meant to cover pine nuts, so this was an intended gap the 2022 split had quietly opened, making roughly \$20M/year of chilgoza exports invisible from 2022 onward. In the opposite direction, HS 2012 had split 0802.50 into two pistachio codes, and the concordance carried only the two post-2012 codes, never the pre-2012 original, dropping roughly \$2.6M/year of pistachio imports for 2003–2011. Both were closed the same way as citrus: verified for non-overlap, then the retired code kept alongside its successors. This also corrected an earlier, over-confident conclusion in this project’s own notes that citrus had been the *only* commodity affected — that conclusion came from a test that only checked the 2017 boundary and worked at the whole-commodity level, which can’t see a gap inside Nuts because almond production dominates the total enough to hide a missing pine-nut or pistachio component.

3.3 The pre-2012 hole: the single largest correction

Running that same sibling test across every revision boundary and every heading the concordance already queries surfaced something considerably bigger: the concordance had been built entirely from post-2012 codes, and the HS 2012 revision had restructured most of the agricultural tariff schedule, so for 2003 through 2011 the modern codes queried by this project simply returned zero across roughly fifteen commodities, while the real trade sat on retired codes nobody had queried. This landed squarely on FABLE’s own calibration years (2000, 2005, 2010). **Wheat** was the worst-affected: for 2003–2011 combined, the pre-fix data captured only \$911.7M of a real \$3,157.5M, and for 2008 alone — the global food-price-crisis year — it captured just \$8.7M of a true \$1.67 billion, missing the 3.7 million tonnes Pakistan actually imported entirely. Sixteen retired codes were added across twelve commodities, restoring \$4.92 billion of previously invisible trade. Every retired code was checked against its *own* successor codes, not its wider tariff heading — a heading-level check gives false positives, since raw cane sugar’s retired code looks like it coexists with modern codes purely because refined sugar, a different product on a different code in the same heading, happens to be live in the same years. Two judgement calls: a pre-2012 banana code that also covers plantains in one line was assigned wholly to Banana, since plantains aren’t grown in Pakistan and post-2012 data shows the heading is over 95% banana anyway (the possible ~4% plantain contamination is noted, not estimated away); and a pure-bred cattle-breeding code was left out, since breeding stock isn’t a food-supply flow.

3.4 Rapeseed, 2000–2001

A final check, working the comparison in reverse — what does FAOSTAT report that the trade dataset shows nothing for at all — found one more gap of the same class at the earlier HS 2002 boundary: FAOSTAT reported roughly \$109M and \$80M of Pakistani rapeseed imports for 2000 and 2001, against “no data” on the trade side. HS 2002 had split the single pre-2002 rapeseed code into two modern codes, and the concordance only ever carried the modern pair. Since 2000 is a FABLE calibration year, this mattered more than an ordinary early-year gap. Verified and fixed the same way: the retired code and its two successors never overlap in time.

3.5 Other concordance fixes found later

Cotton lint’s export total had been dominated by cotton waste, not lint. HS heading 5202 — cotton yarn waste, garnetted stock waste, other spinning/weaving waste — is a textile manufacturing byproduct, not something a grower or ginner produces, and had no business being summed into a primary-product commodity. It nonetheless accounted for 75–80% of Cotton lint’s reported export total in recent years, inflating official exports two to five times over. It survived every earlier audit because it doesn’t look wrong from the inside — no other commodity

claims the codes, so the double-count check never flags it, and the codes return real, non-zero trade every year. Caught by noticing Cotton lint’s FAOSTAT mismatch was unusually large and one-directional (badly overstating exports, barely at all on imports). Dropping the three waste codes brought the 2022 export total to \$17.19M against FAOSTAT’s own \$16.28M, versus a 320% overstatement beforehand — a real correction to the trade dataset’s own totals, not just to the FAOSTAT comparison.

A tariff code labelled “pepper” was actually chillies. HS 0904.21 had been described as “pepper, dried/crushed/ground.” Comtrade’s heading 0904 actually holds two unrelated spices — true pepper on 0904.11/0904.12, chillies on 0904.21/0904.22 — and 0904.21 sits in the chilli half. Nothing had been miscounted, only mislabelled; corrected as a label-only change. Real pepper remains uncovered in every era despite roughly \$500M/year of imports — a scope gap, not a bug, left for the mentor.

Two coverage gaps closed in the final reconciliation pass. An end-to-end check of the trade, concordance and validation deliverables against each other found two more real coverage gaps. Goat & Lamb was missing 0204.21 (adult-sheep/mutton carcasses, fresh/chilled), the single largest sheep-meat export code Pakistan uses (\$8–31M/year) — Pakistan exports predominantly mutton, not lamb, so this one code accounted for most of a ~40% undercount against FAOSTAT. Cotton (seed) was missing 1207.29 (cottonseed for crushing/feed, distinct from seed for sowing). Both were added at factor 1.00, matching their sibling codes; the fix validated itself immediately — after adding them, Goat & Lamb’s export figures matched FAOSTAT’s own published totals to within a few hundred dollars in multiple years.

4 Aggregating Trade into FABLE Commodities

Raw HS-code data is rolled up into one row per FABLE commodity, per year, per flow, in million tonnes; trade value is summed in US dollars with no factor applied, since a dollar figure needs no weight correction. CONVERT rows have their factor applied to quantity only, sourced or not — a partial estimate still beats treating a converted product as the primary form, and every such case is flagged.

Live-animal trade is handled by checking each row’s own reported unit (kilograms versus a head-count), not a hard-coded code list: in the 2000–2002 partner-mirror years live sheep are reported in kilograms rather than as a head-count, so a hard-coded exclusion would have wrongly skipped convertible data for those years.

4.1 Comtrade source-data anomalies

A pass looking for an implied unit price outside any plausible range, combined with an outlier relative to that code’s own historical median, found ten candidate rows, all tagged as Pakistan’s own direct reporter data (meaning any error originates in Pakistan’s customs submission, not this pipeline). The clearest case: Apple’s 2016 import quantity, 60,078,115,076 kg, valued at only \$34.36M — an implied price of \$0.57/tonne against apples’ real \$300–1,500/tonne range, and a quantity larger than roughly two-thirds of the world’s entire annual apple production. The stored figure carries the same leading digits as a plausible 60,078,115 kg, fitting neatly on the trend between 2015 and 2017, with three extra digits appended — a decimal-point or unit error at the original submission, not a coincidence. Corrected by dividing the reported quantity by 1,000.

Nine further rows showed the same signature — rice-paddy imports (2006, 2008), sugar-raw exports (2015, two codes), jute trade (2009, 2010), oats exports (2022), corn/maize imports (2003, 2004) — but none had Apple’s unambiguous resolution: dividing several of these by 1,000 implies an equally absurd price the other way (rice above \$300/kg, for one). These were briefly excluded and then, after review, deliberately reverted to Comtrade’s original figures —

not because they were judged correct, but because excluding a value on suspicion without a resolvable fix risks discarding real data on a guess. Total import/export value across the dataset is unchanged to the cent throughout this episode, since only quantities were ever touched.

5 Joining Production and Trade

5.1 The join-key bug

An initial join matched production and trade records on commodity *name*. It matched only 37 of 42 production commodities, because Excel forbids “/” in a worksheet name, so the production workbook spells five commodities differently from the trade concordance (“Corn - Maize” against “Corn / Maize”, and the same for Gram/Chickpea, Lentil/Beans, Orange (Citrus), and Goat & Lamb (Mutton)). A name-based join drops any commodity whose two spellings don’t match character for character — here, roughly \$1.09 billion of trade, including Maize, Pakistan’s second-largest export by volume, with no error raised. Fixed by adding an explicit **fable_number** column to both tables and joining on that instead. Joining on the number surfaced one more wrinkle: FABLE commodity 37, “Other pulses,” is one production commodity but three trade commodities (Other pulses, Guarseed, Guar gum). Decision: sum all three, since Pakistan is a globally significant guar exporter and excluding guar gum would drop a real export for no good reason.

5.2 Fiscal years versus calendar years

Production is reported by Pakistani fiscal year (July–June, labelled by its starting year), trade by calendar year. Each fiscal year maps to its *first* calendar year (2000-01 joins calendar-year 2000), for three reasons: it matches FAOSTAT’s own convention for Pakistani crop years, which matters because FAOSTAT is the baseline being validated against; it’s the only mapping under which FABLE’s calibration years (2000, 2005, 2010) land on production rows that exist; and it has a physical basis, since the crop year opens in July of its first calendar year and the kharif harvest, carrying the bulk of the season’s tonnage, is gathered that same autumn. Calendar-year 2025 trade has no production counterpart under this mapping and drops out of the joined table — accepted rather than worked around, since a weighted split across the two overlapping fiscal years would need a monthly breakdown Comtrade doesn’t publish.

5.3 A data-type mismatch caught before it could cause harm

A pre-build check, run specifically because the join key had already caused one silent failure, found a second: production’s **fable_number** column was stored as an integer, trade’s as a string. A join on two columns of different types returns zero matched rows, with no error — the same failure shape one level deeper. A written join specification from an earlier session had already covered the choice of key and the year mapping in detail, but hadn’t anticipated the two sides might not share a data type. Lesson: even a carefully specified join deserves an actual dtype check before being trusted.

5.4 Balance methodology and output

The balance (Production + Imports – Exports) is computed only when all three values are present; a missing term is never assumed to be zero. The deliverable presents a complete 76-commodity by 26-year grid rather than only the rows where both sources have data, so the production-side gap (34 commodities with no series) and trade-side gap (12 with no data) are visible in the table itself. All twelve trade-absent commodities are also production-absent, and

every one is a crop Pakistan genuinely neither grows nor trades — a reassuring consistency check.

The deliverable’s shape was simplified once, after review: the two supplementary non-official rows (Groundnut, Soybean) were dropped from this table (they remain in the standalone aggregated trade file); several tracking columns (fiscal-year label, per-row scope notes, HS-code counts) were removed, since a scope mismatch between a commodity’s production and trade coverage is inherent to only including data that exists; and every weight-based figure was unified to million tonnes, simplifying the join’s own arithmetic in the process.

6 FAOSTAT Validation Decisions

6.1 Sourcing FAOSTAT data

FAOSTAT’s live API was unavailable for the entire session this validation was built in, though the FAOSTAT website itself was reachable — an API-specific outage. The official bulk CSV-download mirror was used instead, filtered to Pakistan’s roughly 77,000 rows and saved locally. Units were verified directly from the data rather than assumed — tonnes for quantity, thousands of US dollars for value — since assuming the wrong unit here would have produced exactly the kind of order-of-magnitude “error” this project spent considerable effort chasing elsewhere.

6.2 Filling the concordance’s FAO-code gaps

Nine composite commodities (Nuts, Other oils, Rapeseed cake, Other oilseed cake, Other pulses, Other citrus, Other fruits, Other vegetables, Other spices) had 70 underlying HS rows with no FAOSTAT item at all. Each was checked individually against FAOSTAT’s real 622-item classification. Sixty were mapped to a genuine item; ten were confirmed, by direct search, to have no FAOSTAT counterpart whatsoever (pine nuts, guar seed, guar gum, garlic, celery, turmeric, Brussels sprouts, a salad-root group) and were left unmapped rather than forced into an approximate bucket. Two more carry a scope caveat: Other Fruits’ combined raspberry/blackberry/mulberry code maps only to FAOSTAT’s raspberry item, which understates that comparison, and Other Spices’ coriander-seed code maps to FAOSTAT’s combined coriander/cumin/anise bucket, which overstates it.

6.3 The conversion-scope gap

Every CONVERT row that already had a FAOSTAT code was found, on inspection, to be pointing at the *same* code as its unconverted, primary-form sibling. Rice’s husked, milled and broken forms — 98.7% of Rice’s own trade — were all compared against FAOSTAT’s raw-paddy item, even though FAOSTAT tracks each as a separate item; the same pattern hit Milk, Sugar raw, Wheat, Barley, Corn, Grape, and Cocoa. This made the FAOSTAT side of these comparisons structurally incomplete, making the correctly converted totals look far larger for reasons that had nothing to do with accuracy. Fixed by giving each processed form its own correct FAOSTAT item and rebuilding the comparison to apply each row’s own conversion factor to the matched FAOSTAT quantity before summing. One case had no clean resolution: cocoa’s defatted and non-defatted paste both map to FAOSTAT’s single combined item, so the two factors were averaged instead of picking one arbitrarily. Effect: Wheat, Rice (paddy), Sugar raw, Milk and Grape moved from mostly-red to mostly-green; Rice (paddy)’s 2023 export comparison moved from an 8.7% mismatch to an exact match.

6.4 Outright wrong FAOSTAT codes

A second check — cross-referencing every FAOSTAT code actually used against FAOSTAT’s own name for that code, rather than trusting the label already stored — found seventeen rows

pointing at an unrelated product: Mung bean pointed at the same code the separate Peas commodity correctly uses; Jute pointed at true hemp; Cotton (seed) pointed at a code that doesn't exist in FAOSTAT's list; Cotton cake pointed at crude rapeseed oil; Butter/Ghee pointed at liver preparations; Raw animal fat pointed at sausages. These are pre-existing errors dating to whenever the FAO-code column was first populated, not something this project's earlier work introduced. Each was corrected; Gram/Chickpea's mismatch rate fell from 80% to 38%, Cotton cake's from 82% to 33%.

6.5 Sibling-item mismatches

Three commodities had several tariff codes covering different sub-forms all compared against one FAOSTAT item corresponding to only one sub-form. Pakistan's citrus exports are overwhelmingly kino (a mandarin), but all four kino codes were compared against FAOSTAT's "Oranges" item rather than its separate "Tangerines, mandarins and clementines" item; corrected, the comparison matches FAOSTAT almost to the dollar in the modern era. Offal's sheep/goat codes were compared against FAOSTAT's cattle-offal item rather than its sheep-offal item, and Coconut's desiccated code against its in-shell item rather than its own desiccated item; both corrected the same way.

6.6 Why early years show more disagreement

Comparable cells run 7–13% green for 2000–2002, 20–35% for 2003–2010, and a stable 68–78% from 2011 onward, for two distinct, verifiable reasons. For 2000–2002, the trade dataset itself is a partner-mirror reconstruction, since Pakistan filed nothing with Comtrade those years — a genuine limitation of the early-year trade data, not evidence FAOSTAT is wrong. For 2003–2010, FAOSTAT's own data-quality flags show 20–35% of Pakistan's trade rows in that window are marked estimated, imputed, or drawn from an international rather than national source, falling to ~2% from 2014 onward — direct evidence that FAOSTAT itself relied heavily on estimation for this period rather than Pakistan's actual customs submissions, held directly via Comtrade. From 2011 on, FAOSTAT increasingly sources from Comtrade too, which is why the two converge — Potato, with no conversion complications, matches FAOSTAT to the dollar in nearly every year from 2011 onward. The conclusion: the early-year gap should not be closed by altering the more reliable Comtrade-derived figures to match FAOSTAT's estimates for that period, since doing so would defeat the entire purpose of validating against FAOSTAT rather than being fitted to it.

A smaller set of mismatches are genuine, permanent scope differences rather than errors: Mung bean and the Mash-bean portion of Lentil/Beans have no FAOSTAT-specific item and compare against a broad "dry beans" bucket; Other Vegetables includes garlic, which Pakistan imports in real volume but FAOSTAT tracks under no item at all; and Rubber shows a persistent overstatement on the trade-dataset side for a crop Pakistan doesn't produce domestically, most plausibly FAOSTAT's own incompleteness for a non-produced industrial input.

7 Full Findings Table

Finding	Status	Impact
Pre-2012 HS-revision hole across ~15 commodities, headlined by Wheat	Fixed	\$4.92 billion of trade restored across 12 commodities for 2003–2011; Wheat's 2008 imports alone recovered from \$8.7M to \$1.67bn.
Citrus (kino) tariff code retired at HS 2017	Fixed	Orange/Citrus 2023 exports recovered from \$2.82M to \$76.77M; restored ~300,000 t/year of Pakistan's largest fruit export.

Finding	Status	Impact
Nuts: pine-nut code retired at HS 2022; pistachio code split at HS 2012	Fixed	Restored ~\$20M/year of pine-nut exports and ~\$2.6M/year of pistachio imports.
Rapeseed pre-2002 tariff code never added	Fixed	Recovered ~\$78M (2000) and ~\$74M (2001) of imports, at a FABLE calibration year.
Cotton lint export total dominated by manufacturing-waste tariff codes	Fixed	Corrected a 2–5× overstatement; 2022 export figure moved to \$17.19M, matching FAOSTAT within 5.6%.
Guar gum summed at raw-gum weight instead of seed-equivalent	Fixed	Factor corrected from 1.00 to 3.45 once a sourced seed-to-gum yield was found.
Sunflower, Jute, Other Oils left in raw tonnes instead of thousands of tonnes	Fixed	Corrected an up-to-1000× overstatement.
Cotton lint in bales, Eggs by count, not weight	Fixed	Converted to the workbook’s standard unit (170 kg/bale; 47 g/egg).
24 FAOSTAT-comparison rows compared a converted total against an unconverted item	Fixed	Wheat, Rice, Milk, Sugar raw, Grape moved from mostly-red to mostly-green.
17 FAOSTAT codes pointed at an unrelated product	Fixed	Gram/Chickpea’s mismatch fell from 80% to 38%; Cotton cake’s from 82% to 33%.
Production–trade join on name dropped 5 of 42 commodities	Fixed	Recovered ~\$1.09 billion of trade, including Maize.
Apple’s 2016 import quantity overstated by exactly 1000×	Fixed	Corrected from 60.08 million tonnes to 60,078 tonnes.
Peas, Punjab, 2017-18: reported figure is ~10× its neighbours	Flagged, kept as-is	Source’s own national total carries the identical error; kept as reported.
9 further Comtrade rows show an implausible price with no unambiguous fix	Flagged, kept as-is	Left as Pakistan’s own reported figures rather than excluded on suspicion.
Sugar raw, 2023-24: entire table is a verbatim repeat of 2021-22	Flagged, kept as-is	Kept as published, flagged in the sheet’s notes.
Livestock and Milk, 2011-12 through 2017-18	Limitation	Both available ASP editions print only the most recent 3–5 years.
Cotton (seed), Butter/Ghee, Other Citrus have no production series	Limitation	Checked against every source obtained; genuinely absent.
Real pepper uncovered in every era despite ~\$500M/year of imports	Limitation	A scope decision, flagged for the mentor rather than added unilaterally.

8 Reproducibility Conventions

Every one-time fix script asserts the exact pre-fix state of a row before writing to it, so an accidental re-run can’t misapply a fix to data that has since changed; several refuse a second run outright. Structural edits to a workbook with merged cells — the concordance table, most notably — rebuild the sheet from scratch rather than inserting or deleting rows in place, after an early in-place insertion once corrupted the file’s merged banner cells and destroyed unrelated data elsewhere.

Each of the project’s seven Excel deliverables ships with a small, dependency-free Python script that reproduces that exact file byte for byte, by embedding the finished workbook’s bytes as a base64 string; every one was verified via SHA-256 hash comparison to reproduce its target file exactly. A structured decision log (`qa_log.csv`) records every non-obvious decision with its external citation, appended to throughout the project rather than written up after the fact.

9 Limitations

This dataset is only as complete as Pakistan’s own published statistics allow, and the gaps below were checked and confirmed rather than assumed.

Production side. Livestock products — milk, beef, mutton, poultry, eggs and animal fat — have no provincial breakdown in any source checked; every livestock figure is a national total. Livestock and milk data for 2011-12 through 2017-18 remain unfillable with sources on hand, since both available ASP editions print only the three to five most recent years in their livestock tables; 2024-25 is not yet covered by any published edition; and dates production before 2004 has no table in the earliest PBS source available. Cotton (seed), Butter/Ghee, and Other Citrus have no production series at all, confirmed absent across every ASP, PBS and FVC source obtained.

Trade and validation side. A small number of conversion factors — rice flour and one catch-all milk-powder category, in particular — are honest estimates rather than fully verified figures, checked against every reference this project draws on and found genuinely uncovered; they are labelled as such. A handful of real Pakistani trade flows sit outside FABLE’s own commodity classification entirely — real pepper, kidney beans, palm kernel cake, copra, and cardamom/cumin all represent genuine, sizeable trade with no existing FABLE slot to receive it, and adding one is a scope decision rather than something resolved unilaterally here.